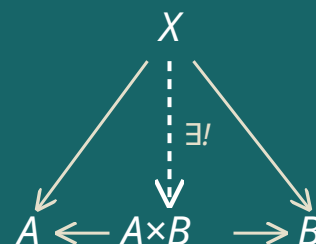


AS.110.615, Johns Hopkins University

Algebraic Topology I

Fall Semester 2026



Lectures — MW 1:30–2:45 pm,
Bloomberg 276 (Homewood
campus)

Instructor — David Gepner,
Krieger 410; office hours TBD, and
by appointment

About the Course

This is the first semester of the graduate algebraic topology sequence. The guiding idea of the subject is to attach algebraic invariants — groups, modules, chain complexes — to topological spaces in a functorial way, and to use these invariants to distinguish spaces and to prove theorems inaccessible by point-set methods alone.

Prerequisites. The equivalent of one semester each of graduate abstract algebra and real analysis, including point-set topology. Prior exposure to the fundamental group (e.g., at the level of AS.110.413) is helpful but not assumed. Comfort with modules over a ring and basic categorical language will be assumed after the first two weeks.

Lectures

Dates	Topic
1 Aug 31 & Sep 2	<u>Additive and Abelian Categories</u>
2 Sep 9	<u>Exact Sequences and Derived Categories</u>
3 Sep 14 & 16	<u>Tensor Products, Tor, Ext, and Simplicial Chains</u>
4 Sep 21 & 23	<u>Simplicial Sets, Spaces, and Singular Chains</u>

Dates	Topic
5 Sep 28 & 30	<u>Generalized Homology and the Eilenberg-Steenrod Axioms</u>
6 Oct 5 & 7	<u>Homotopy Invariance and First Computations</u>
7 Oct 12 & 14	<u>Homotopy Pushouts and Mayer-Vietoris</u>
8	<u>Homotopy Pushouts and Singular Chains, Part I</u>

Syllabus

Part I — Categories and Homological Algebra

Week 1	August 31 and September 2. Categories, functors, natural transformations, limits, colimits, and adjunctions. Additive and abelian categories; kernels and cokernels.
Week 2	September 9. Chain complexes, chain homotopies, and homology. Exact sequences, the snake lemma, and long exact sequences. Mapping cones; homotopy and derived categories.
Week 3	September 14 and 16. Tensor products, resolutions, derived functors, Tor, and Ext. Extensions and noncommutative examples. Simplicial chains and universal coefficients.
Week 4	September 21 and 23. The simplex category; simplicial sets, faces, and degeneracies. Geometric realization, singular simplicial sets, and their adjunction. Simplicial abelian groups, singular chains, and the statement of Dold-Kan.

Part II — Singular Homology

Week 5	September 28 and 30. Generalized homology and the Eilenberg-Steenrod axioms in Mayer-Vietoris form. The homotopy category and homotopy invariant functors. Beginning of the proof of homotopy invariance of singular homology.
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Week 6	October 5 and 7. Homotopy invariance of singular homology; homotopy pushout squares in Top and Ch; double mapping cylinders, mapping cones, relative homology, and reduced homology.
Week 7	October 12 and 14. Homotopy pushouts; relative and reduced homology. Preservation of homotopy pushouts by singular chains with coefficients in an abelian group A . The long exact sequence in homology of a homotopy pushout of chain complexes. Mayer-Vietoris and excision; homology of spheres, tori, and complex projective spaces.
Week 8	October 19 and 21. CW complexes and cellular homology; comparison with singular homology. Attaching maps, degree, and Euler characteristic. Universal coefficients and a survey of the Künneth theorem. Applications to the Brouwer fixed point theorem, invariance of dimension, and invariance of domain.

Part III — Homotopy Theory and the Fundamental Group

Week 9	October 26 and 28. Homotopy groups, basepoints, and the action of the fundamental group. Statements of the Whitehead and Hurewicz theorems; proofs deferred to Math 616.
Week 10	November 2 and 4. Fibrations and the homotopy lifting property; Serre fibrations and fiber bundles. The long exact sequence of a fibration, the path-loop fibration, and the Hopf fibration; computation of the third homotopy group of the 2-sphere.
Week 11	November 9 and 11. The fundamental group: the circle via lifting, free products, and the Seifert-van Kampen theorem. Applications to the fundamental theorem of algebra and the low-dimensional Borsuk-Ulam theorem.
Week 12	November 16 and 18. Covering spaces: lifting criteria, classification by subgroups of the fundamental group, universal covers, deck transformations, and the Galois correspondence. Subgroups of free groups; preview of Math 616. Remaining class time is reserved for review and project preparation; the Lefschetz fixed point theorem is included if time permits.

Final Project Presentations

**Nov 30-
Dec 9** In-class final project presentations. No new lecture material.

The schedule from October 12 onward is provisional. Thanksgiving break: no classes during the university break.

Grading

Your course grade is based entirely on class participation/attendance and a final project. There are no exams.

- 1. Participation and attendance (10%).** This is a lecture course, but a conversational one: attend regularly, ask and answer questions, and occasionally present an argument at the board when invited. The participation grade reflects consistent, good-faith engagement over the semester — not the number of times you speak, and certainly not whether your questions are "good enough." If you must miss more than an occasional class (illness, travel, conflicts), just let me know.
- 2. Final project (90%).** An in-class presentation on a topic extending the course, to take place during the last two weeks of class (Nov 30 – Dec 9). There is no written paper. Suitable topics include: the Dold–Kan correspondence in full; model structures on simplicial sets; group cohomology via derived functors; the Hurewicz theorem; classifying spaces and principal bundles; Poincaré duality; the Lefschetz fixed point theorem; covering space theory from the Galois-theoretic viewpoint; braid groups and configuration spaces. Topic proposal due mid-October; optional draft of your presentation materials early November. Graded on mathematical correctness, depth of engagement with the material, and clarity of exposition — not originality of research.
- 3. Homework (assigned, not counted toward the grade).** One problem set per week, graded for your benefit only and returned with comments. Collaboration is encouraged; write up solutions in your own words.

Homework

Week	Dates	Topic
1	Aug 31 & Sep 2	<u>Categories and Chain Complexes</u>
2	Sep 9	<u>Homotopies, Exact Sequences, and Cones</u>
3	Sep 14 & 16	<u>Derived Functors, Tor, and Ext</u>
4	Sep 21 & 23	<u>Simplicial Sets and Singular Chains</u>
5	Sep 30	<u>Generalized Homology and the Eilenberg-Steenrod Axioms</u>
6	Oct 5 & 7	<u>Homotopy Invariance</u>
7	Oct 12 & 14	<u>Homotopy Pushouts and Homology</u>

Important Dates

Date

Mon, Aug 31	First class
Mid-October	Project topic proposal due
Early Nov	Optional draft of presentation materials
Late Nov	Thanksgiving break
Nov 30 – Dec 9	Final project presentations (last two weeks of class); last class Wed, Dec 9

Policies

1. **Collaboration and integrity.** Discussion of homework with classmates is encouraged. The final project must be your own exposition; all sources (including AI tools, if used) must be cited. All work is subject to the university's graduate academic misconduct policies.

2. **AI use.** Prohibited on homework: the problem sets exist entirely for your benefit, so outsourcing them defeats the purpose. Permitted for everything else, subject to the citation requirement for the final project.
3. **Devices in class.** Discouraged. If your attention drifts, work on something productive instead — a lit screen makes it harder for everyone nearby to pay attention.
4. **Accommodations.** Students requiring accommodations should register with [Student Disability Services](#) and contact me early in the semester.

Texts

No single text covers the course in the order we follow. Primary references, all freely or electronically available:

1. A. Hatcher, *Algebraic Topology* — [free from the author's page](#). Main reference for homology, the fundamental group, and covering spaces.
2. C. Weibel, *An Introduction to Homological Algebra* — [Cambridge University Press](#); main reference for Part I.
3. S. Gelfand and Y. Manin, *Methods of Homological Algebra* — [SpringerLink](#); for derived and triangulated categories.
4. P. Goerss and J. F. Jardine, *Simplicial Homotopy Theory* — [SpringerLink](#); main reference for simplicial sets.
5. J. P. May, *A Concise Course in Algebraic Topology* — [free PDF from the author's page](#); secondary reference throughout, especially for fibrations.

AS.110.615, Department of Mathematics, Johns Hopkins University. Questions about the course: gepner@jhu.edu. Page maintained by the instructor; last updated October 7, 2026.